Dr .SNS RAJALAKSHMI COLLEGE OF ARTS & SCIENCE (AUTONOMOUS)

Accredited by NAAC (Cycle-III) with 'A+'
Grade Affiliated to Bharathiyar University
Coimbatore-641049

ESTD : 1999

DEPARTMENT OF COMPUTER SCIENCE WITH CYBER SECURITY

II- B.SC CS WITH CYBER SECURITY

Certified That This Bonafide Record Work Done By

NAME : ……..

REG.NO : ……...

BATCH :2022-2025

23UCU559- CORE LABORATORY - 4: RELATIONAL DATABASE MANAGEMENT SYSTEM

STAFF INCHARGE **SIGNATURE OF HOD**

Submitted for B.sc Degree IV Semester

Practical Examination held on……

INTERNAL EXAMINER **EXTERNAL EXAMIN**

INDEX

S.No	Date	Particulars	Pg.No	Signature

01.DML AND TCL COMMANDS

Aim:

Algorithm:

Coding:

```
CREATE TABLE Students(
student_id INT,
student_name VARCHAR(20),
department VARCHAR(20),
age INT
);
insert into Students values(1,'ajith','bsc',20);
insert into Students values(2,'abinaya','bsc',20);
insert into Students values(3,'deva','bcom',19);
insert into Students values(4,'kaviya','bca',20);
select *FROM Students;
ALTER TABLE Students ADD MARKS INT;
UPDATE Students SET MARKS=100;
SELECT *FROM Students;
commit;
delete from Students where student_id=1;
select *FROM Students;
rollback;
select *FROM Students;
```

Output:

```
SQL> select *from Students;

STUDENT_ID STUDENT_NAME         DEPARTMENT                   AGE
---------- -------------------- -------------------- ----------
         1 ajith                bsc                          20
         2 abinaya              bsc                          20
         3 deva                 bcom                         19
         4 kaviya               bca                          20

SQL> ALTER TABLE Students ADD MARKS INT;

Table altered.

SQL> UPDATE Students SET MARKS=100;

4 rows updated.

SQL> SELECT *FROM Students;

STUDENT_ID STUDENT_NAME         DEPARTMENT                   AGE      MARKS
---------- -------------------- -------------------- ---------- ----------
         1 ajith                bsc                          20        100
         2 abinaya              bsc                          20        100
         3 deva                 bcom                         19        100
         4 kaviya               bca                          20        100

SQL> commit;

Commit complete.

SQL> delete from Students where student_id=1;

1 row deleted.

SQL> select *FROM Students;

STUDENT_ID STUDENT_NAME         DEPARTMENT                   AGE      MARKS
---------- -------------------- -------------------- ---------- ----------
         2 abinaya              bsc                          20        100
         3 deva                 bcom                         19        100
         4 kaviya               bca                          20        100

SQL> rollback;

Rollback complete.

SQL> select *FROM Students;
```

2. SQL SPECIAL OPERATORS

Aim:

Algorithm:

Coding:

```
CREATE TABLE CUSTOMERS (
  ID INT,  NAME VARCHAR(20) NOT NULL,
  AGE INT NOT NULL,
  ADDRESS CHAR (25),
  SALARY DECIMAL (18, 2),
  PRIMARY KEY (ID)
);
INSERT INTO CUSTOMERS VALUES (1, 'Ramesh', 32, 'Ahmedabad', 2000.00 );
INSERT INTO CUSTOMERS VALUES(2, 'Khilan', 25, 'Delhi', 1500.00 );
INSERT INTO CUSTOMERS VALUES(3, 'Kaushik', 23, 'Kota', 2000.00 );
INSERT INTO CUSTOMERS VALUES(4, 'Chaitali', 25, 'Mumbai', 6500.00 );
INSERT INTO CUSTOMERS VALUES(5, 'Hardik', 27, 'Bhopal', 8500.00 );
INSERT INTO CUSTOMERS VALUES(6, 'Komal', 22, 'Hyderabad', 4500.00 );
INSERT INTO CUSTOMERS VALUES(7, 'Muffy', 24, 'Indore', 10000.00 );
SELECT ID, NAME, SALARY FROM CUSTOMERS
WHERE SALARY > 2000 AND AGE < 25;
SELECT * FROM CUSTOMERS WHERE AGE BETWEEN 20 AND 25;
SELECT * FROM CUSTOMERS
WHERE NAME IN ('Khilan', 'Hardik', 'Muffy');
SELECT * FROM CUSTOMERS WHERE SALARY LIKE '200%';
```

Output:

```
SQL> SELECT ID, NAME, SALARY FROM CUSTOMERS
  2  WHERE SALARY > 2000 AND AGE < 25;

        ID NAME                    SALARY
---------- -------------------- ----------
         6 Komal                      4500
         7 Muffy                     10000

SQL> SELECT * FROM CUSTOMERS WHERE AGE BETWEEN 20 AND 25;

        ID NAME                       AGE ADDRESS                        SALARY
---------- -------------------- ---------- ------------------------- ----------
         2 Khilan                       25 Delhi                           1500
         3 Kaushik                      23 Kota                            2000
         4 Chaitali                     25 Mumbai                          6500
         6 Komal                        22 Hyderabad                       4500
         7 Muffy                        24 Indore                         10000

SQL> SELECT * FROM CUSTOMERS
  2  WHERE NAME IN ('Khilan', 'Hardik', 'Muffy');

        ID NAME                       AGE ADDRESS                        SALARY
---------- -------------------- ---------- ------------------------- ----------
         2 Khilan                       25 Delhi                           1500
         5 Hardik                       27 Bhopal                          8500
         7 Muffy                        24 Indore                         10000

SQL> SELECT * FROM CUSTOMERS WHERE SALARY LIKE '200%';

        ID NAME                       AGE ADDRESS                        SALARY
---------- -------------------- ---------- ------------------------- ----------
         1 Ramesh                       32 Ahmedabad                       2000
         3 Kaushik                      23 Kota                            2000
```

3. AGGREATE FUNCTION

Aim:

Algorithm:

Coding:

```
CREATE TABLE CUSTOMERS (
  ID INT,  NAME VARCHAR(20) NOT NULL,
  AGE INT NOT NULL,
  ADDRESS CHAR (25),
  SALARY DECIMAL (18, 2),
  PRIMARY KEY (ID)
);
INSERT INTO CUSTOMERS VALUES (1, 'Ramesh', 32, 'Ahmedabad', 2000.00 );
INSERT INTO CUSTOMERS VALUES(2, 'Khilan', 25, 'Delhi', 1500.00 );
INSERT INTO CUSTOMERS VALUES(3, 'Kaushik', 23, 'Kota', 2000.00 );
INSERT INTO CUSTOMERS VALUES(4, 'Chaitali', 25, 'Mumbai', 6500.00 );
INSERT INTO CUSTOMERS VALUES(5, 'Hardik', 27, 'Bhopal', 8500.00 );
INSERT INTO CUSTOMERS VALUES(6, 'Komal', 22, 'Hyderabad', 4500.00 );
INSERT INTO CUSTOMERS VALUES(7, 'Muffy', 24, 'Indore', 10000.00 );
Select * From CUSTOMERS;
SELECT AVG(SALARY) From CUSTOMERS;
SELECT COUNT(*) From CUSTOMERS;
SELECT MAX(SALARY) from CUSTOMERS;
SELECT MIN(SALARY) from CUSTOMERS;
SELECT SUM(SALARY) From CUSTOMERS;
```

Output:

```
SQL> Select * From CUSTOMERS;

        ID NAME                          AGE ADDRESS                        SALARY
---------- -------------------- ---------- ------------------------- ----------
         1 Ramesh                        32 Ahmedabad                       2000
         2 Khilan                        25 Delhi                           1500
         3 Kaushik                       23 Kota                            2000
         4 Chaitali                      25 Mumbai                          6500
         5 Hardik                        27 Bhopal                          8500
         6 Komal                         22 Hyderabad                       4500
         7 Muffy                         24 Indore                         10000

7 rows selected.

SQL> SELECT AVG(SALARY) From CUSTOMERS;

AVG(SALARY)
-----------
       5000

SQL> SELECT COUNT(*) From CUSTOMERS;

  COUNT(*)
----------
         7

SQL> SELECT MAX(SALARY) from CUSTOMERS;

MAX(SALARY)
-----------
      10000

SQL> SELECT MIN(SALARY) from CUSTOMERS;

MIN(SALARY)
-----------
       1500

SQL> SELECT SUM(SALARY) From CUSTOMERS;

SUM(SALARY)
-----------
      35000
```

4.FUNCTIONS

Aim:

Algorithm:

Coding:

```
CREATE TABLE Ex3(
item_id INTEGER,
item_name VARCHAR(30),
checkin_date DATE,
item_price NUMBER(10,3),
branch_name VARCHAR(200),
consumer_name VARCHAR(20),
place VARCHAR(20)
);
INSERT INTO Ex3 VALUES(12101,'MacBook',TO_DATE('2023-07-28','YYYY-MM-
DD'),350000.00,'
Apple Store','RAJA','London');
INSERT INTO Ex3 VALUES(12101,'Watch',TO_DATE('2024-02-20','YYYY-MM-
DD'),4000.00,'Titan','Vijay','London');
INSERT INTO Ex3 VALUES(12101,'Book',TO_DATE('2024-03-11','YYYY-MM-DD'),500.00,'Tamil
Store','Ajith','India');
INSERT INTO Ex3 VALUES(12101,'Gold Chain',TO_DATE('2024-01-20','YYYY-MM-
DD'),55000.00,'Joyalukkas','Leo','Kasmir');
INSERT INTO Ex3 VALUES(12101,'Diamond Ring',TO_DATE('2024-01-31','YYYY-MM-
DD'),5500000.00,'Peter Store','Gwen','New York');
SELECT * FROM Ex3 ;
/*string*/
select upper(item_name) from Ex3;
select initcap(item_name) from Ex3 ;
select trim (both 'e' from item_name) from Ex3 ;
/*date*/
select add_months(TO_DATE('2024-02-20','YYYY-MM-DD'), 3)from dual;
select last_day (TO_DATE('2024-03-11','YYYY-MM-DD')) from dual;
/*number*/
select max(item_price), min(item_price),round( avg(item_price),1), sum(item_price) from Ex3 ;
select min(item_name) , max(item_name) from Ex3 ;
select count(place) from Ex3 ;
/*conversion*/
select to_char(sysdate, 'dd ' )from dual;
select to_char(sysdate, 'ddth' )from dual;
select to_char(sysdate, 'ddspth ' )from dual;
```

Output:

```
SQL> SELECT * FROM Ex3 ;

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
     12101 MacBook                        28-JUL-23     350000
Apple
Store
RAJA                 London

     12101 Watch                          20-FEB-24       4000
Titan

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
Vijay                London

     12101 Book                           11-MAR-24        500
Tamil Store
Ajith                India

     12101 Gold Chain                     20-JAN-24      55000

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
Joyalukkas
Leo                  Kasmir

     12101 Diamond Ring                   31-JAN-24    5500000
Peter Store
Gwen                 New York
```

(1)

```
SQL> /*string*/

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
     12101 MacBook                        28-JUL-23     350000
Apple
Store
RAJA                 London

     12101 Watch                          20-FEB-24       4000
Titan

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
Vijay                London

     12101 Book                           11-MAR-24        500
Tamil Store
Ajith                India

     12101 Gold Chain                     20-JAN-24      55000

   ITEM_ID ITEM_NAME                      CHECKIN_D ITEM_PRICE
---------- ------------------------------ --------- ----------
BRANCH_NAME
--------------------------------------------------------------------
CONSUMER_NAME        PLACE
-------------------- --------------------
Joyalukkas
Leo                  Kasmir

     12101 Diamond Ring                   31-JAN-24    5500000
Peter Store
Gwen                 New York
```

(2)

```
SQL> select upper(item_name) from Ex3;

UPPER(ITEM_NAME)
------------------------------
MACBOOK
WATCH
BOOK
GOLD CHAIN
DIAMOND RING

SQL> select initcap(item_name) from Ex3 ;

INITCAP(ITEM_NAME)
------------------------------
Macbook
Watch
Book
Gold Chain
Diamond Ring

SQL>  select trim (both 'e' from item_name) from Ex3 ;

TRIM(BOTH'E'FROMITEM_NAME)
------------------------------
MacBook
Watch
Book
Gold Chain
Diamond Ring

SQL> /*date*/

TRIM(BOTH'E'FROMITEM_NAME)
------------------------------
MacBook
Watch
Book
Gold Chain
Diamond Ring

SQL> select add_months(TO_DATE('2024-02-20','YYYY-MM-DD'), 3)from dual;

ADD_MONTH
---------
20-MAY-24
```

(3)

```
SQL>  /*number*/

LAST_DAY(
---------
31-MAR-24

SQL> select max(item_price), min(item_price),round( avg(item_price),1), sum(item_price) from Ex3 ;

MAX(ITEM_PRICE) MIN(ITEM_PRICE) ROUND(AVG(ITEM_PRICE),1) SUM(ITEM_PRICE)
--------------- --------------- ------------------------ ---------------
        5500000             500                  1181900         5909500

SQL> select min(item_name) , max(item_name) from Ex3 ;

MIN(ITEM_NAME)                 MAX(ITEM_NAME)
------------------------------ ------------------------------
Book                           Watch

SQL> select count(place) from Ex3 ;

COUNT(PLACE)
------------
           5

SQL>
SQL> /*conversion*/

COUNT(PLACE)
------------
           5

SQL> select to_char(sysdate, 'dd ' )from dual;

TO_
---
15

SQL> select to_char(sysdate, 'ddth' )from dual;

TO_C
----
15th

SQL>  select to_char(sysdate, 'ddspth ' )from dual;

TO_CHAR(SYSDATE
---------------
fifteenth
```

(4)

5.SQL JOINS

Aim:

Algorithm:

Coding:

```
create table department(
 deptid number(2) constraint department_pk primary key,
 deptname varchar(20),
 location varchar(10));
insert into department values (1,'cs','cbe');
 insert into department values(2,'cs','cbe');
insert into department values(3,'it','che');
insert into department values(4,'ct','erd');
create table employees  (
empid NUMBER(4) CONSTRAINT employees  _pk PRIMARY KEY,
empname varchar(10),
deptid NUMBER(2) CONSTRAINT employees  _deptid_pk REFERENCES department(deptid)
);
insert into employees   values(101,'aa',1);
 insert into employees  values(102,'bb',2);
 insert into employees  values(103, 'cc',1);
insert into employees  values(104,'dd',3);
select d.deptname, e.empname from department d join employees  e on d.deptid=e.deptid where d.deptid>2 order
by d.deptname;
select d.deptname, e.empname from department d left outer join employees  e on d.deptid=e.deptid where
d.deptid>2 order by d.deptname;
select d.deptname, e.empname from department d right outer join employees  e on d.deptid=e.deptid where
d.deptid>2 order by d.deptname;
 select d.deptname, e.empname from department d full outer join employees  e on d.deptid=e.deptid where
d.deptid>2 order by d.deptname;
select d.deptname, e.empname from department d cross join employees  e where
d.deptid>2 order by d.deptname;
```

Output:

```
SQL> select d.deptname, e.empname from department d join employees  e on d.deptid=e.deptid where d.d
eptid>2 order by d.deptname;

DEPTNAME             EMPNAME
-------------------- ----------
it                   dd

SQL> select d.deptname, e.empname from department d left outer join employees  e on d.deptid=e.depti
d where d.deptid>2 order by d.deptname;

DEPTNAME             EMPNAME
-------------------- ----------
ct
it                   dd

SQL> select d.deptname, e.empname from department d right outer join employees  e on d.deptid=e.dept
id where d.deptid>2 order by d.deptname;

DEPTNAME             EMPNAME
-------------------- ----------
it                   dd

SQL>  select d.deptname, e.empname from department d full outer join employees  e on d.deptid=e.dept
id where d.deptid>2 order by d.deptname;

DEPTNAME             EMPNAME
-------------------- ----------
ct
it                   dd

SQL>  select d.deptname, e.empname from department d cross join employees  e where d.deptid>2 order
by d.deptname;

DEPTNAME             EMPNAME
-------------------- ----------
ct                   cc
ct                   bb
ct                   aa
ct                   dd
it                   cc
it                   bb
it                   aa
it                   dd

8 rows selected.
```

6.SUB QUIRES

Aim:

Algorithm:

Coding:

```
CREATE TABLE EX6(
id number(10),
fname varchar2(15),
lname varchar2(15),
email varchar2(30),
jodid varchar2(10),
salary number(10),
mid number(5),
deptid number(5)
);
insert into EX6 values (761,'deepika','balu','deepabalu2005@gmail.com','ap',70000,123,12);
 insert into EX6 values (763,'madhu','hasini','medhunhasini2014@gmail.com','ap',70000,124,12);
insert into EX6 values (765,'jeen','mersaline','jeenmersaline2002@gmail.com','HOD',100000,null,12);
 insert into EX6 values (767,'sambath','kumar','sambathkumar2004@gmail.com','HOD',100000,null,15);
 insert into EX6 values (769,'nandhini','mam','nandhini2011@gmail.com','ap',60000,125,15);
 insert into EX6 values (721,'siva','kumar','sivakumar2016@gmail.com','ap',65000,126,15);
 select (fname||''||lname) as name,salary,deptid,jodid from EX6 where jodid=(select jodid from EX6
where id=769);
select id,fname||' '||lname from EX6 where salary > (select avg(salary) from EX6);
select id,fname||' '||lname name,salary from EX6 where mid = (select id from EX6 where fname = 'jeen');
select * from EX6 where salary=(select salary from EX6 where fname='deepika');
```

Output:

```
SQL>  select (fname||''||lname) as name,salary,deptid,jodid from EX6 where jodid=(select jodid from
EX6 where id=769);

NAME                              SALARY     DEPTID JODID
------------------------------ ---------- ---------- ----------
deepikabalu                        70000         12 ap
madhuhasini                        70000         12 ap
nandhinimam                        60000         15 ap
sivakumar                          65000         15 ap

SQL> select id,fname||' '||lname from EX6 where salary > (select avg(salary) from EX6);

        ID FNAME||''||LNAME
---------- ------------------------------
       765 jeen mersaline
       767 sambath kumar

SQL> select id,fname||' '||lname name,salary from EX6 where mid = (select id from EX6 where fname =
'jeen');

no rows selected

SQL> select * from EX6 where salary=(select salary from EX6 where fname='deepika');

        ID FNAME           LNAME           EMAIL
---------- --------------- --------------- ------------------------------
JODID          SALARY        MID     DEPTID
---------- ---------- ---------- ----------
       761 deepika         balu            deepabalu2005@gmail.com
ap               70000        123         12

       763 madhu           hasini          medhunhasini2014@gmail.com
ap               70000        124         12
```

7.DCL COMMANDS

Aim:

Algorithm:

Coding:

```
Create user customer identified by oracle;
Grant connect to customer;
Grant all privileges to customer;
Grant select on customer to ex2;
Revoke select on customer from ex2;
```

Output:

```
User customer created
Grant succeeded
Revoked succeeded
```

8.SEQUENCE AND VIEWS

Aim:

Algorithm:

Coding:

```
CREATE TABLE STUDENT(RNO NUMBER(10),NAME CHAR(20),AGE NUMBER(2),CITY CHAR(10),MARK
S NUMBER(3));
INSERT INTO STUDENT VALUES(1, 'Sasha',17,'Faridabad',80);
INSERT INTO STUDENT VALUES(2,'John',18,'Agra',91);
INSERT INTO STUDENT VALUES(3,'Sara',17,'Hisar',86);
INSERT INTO STUDENT VALUES(4,'Rohan',16,'Faridabad',79);
INSERT INTO STUDENT VALUES(5,'Virat',18,'Delhi',80);
-- Start the transaction
SET AUTOCOMMIT OFF;
-- Inserting data into the STUDENT table
INSERT INTO STUDENT VALUES(6,'Komal',20,'Faridabad',97);
-- Creating savepoint 1
SAVEPOINT save1;
-- Displaying changes after savepoint1
SELECT * FROM STUDENT;
-- Inserting more data into the STUDENT table
INSERT INTO STUDENT VALUES(7,'Preeti',19,'UP',91);
-- Creating savepoint 2
SAVEPOINT save2;
-- Displaying changes after savepoint2
SELECT * FROM STUDENT;
-- Updating marks for a student
UPDATE STUDENT SET MARKS=95 WHERE NAME='Komal';
-- Creating savepoint 3
SAVEPOINT save3;
-- Displaying changes after savepoint3
SELECT * FROM STUDENT;
-- Committing the transaction
COMMIT;
-- Displaying the final changes
SELECT * FROM STUDENT;
-- Disconnect from the database
DISCONNECT;
```

Output:

```
SQL> SET AUTOCOMMIT OFF;
SQL>
SQL> -- Inserting data into the STUDENT table
SQL> INSERT INTO STUDENT VALUES(6,'Komal',20,'Faridabad',97);

1 row created.

SQL> -- Creating savepoint 1
SQL> SAVEPOINT save1;

Savepoint created.

SQL> -- Displaying changes after savepoint1
SQL> SELECT * FROM STUDENT;

       RNO NAME                       AGE CITY            MARKS
---------- -------------------- ---------- ---------- ----------
         1 Sasha                        17 Faridabad          80
         2 John                         18 Agra               91
         3 Sara                         17 Hisar              86
         4 Rohan                        16 Faridabad          79
         5 Virat                        18 Delhi              80
         6 Komal                        20 Faridabad          97

6 rows selected.

SQL>
SQL> -- Inserting more data into the STUDENT table
SQL> INSERT INTO STUDENT VALUES(7,'Preeti',19,'UP',91);

1 row created.

SQL> -- Creating savepoint 2
SQL> SAVEPOINT save2;

Savepoint created.

SQL> -- Displaying changes after savepoint2
SQL> SELECT * FROM STUDENT;

       RNO NAME                       AGE CITY            MARKS
---------- -------------------- ---------- ---------- ----------
         1 Sasha                        17 Faridabad          80
         2 John                         18 Agra               91
         3 Sara                         17 Hisar              86
         4 Rohan                        16 Faridabad          79
         5 Virat                        18 Delhi              80
         6 Komal                        20 Faridabad          97
         7 Preeti                       19 UP                 91


SQL> -- Updating marks for a student
SQL> UPDATE STUDENT SET MARKS=95 WHERE NAME='Komal';

1 row updated.

SQL> -- Creating savepoint 3
SQL> SAVEPOINT save3;

Savepoint created.

SQL> -- Displaying changes after savepoint3
SQL> SELECT * FROM STUDENT;

       RNO NAME                       AGE CITY            MARKS
---------- -------------------- ---------- ---------- ----------
         1 Sasha                        17 Faridabad          80
         2 John                         18 Agra               91
         3 Sara                         17 Hisar              86
         4 Rohan                        16 Faridabad          79
         5 Virat                        18 Delhi              80
         6 Komal                        20 Faridabad          95
         7 Preeti                       19 UP                 91

7 rows selected.

SQL>
SQL> -- Committing the transaction
SQL> COMMIT;

Commit complete.

SQL> -- Displaying the final changes
SQL> SELECT * FROM STUDENT;

       RNO NAME                       AGE CITY            MARKS
---------- -------------------- ---------- ---------- ----------
         1 Sasha                        17 Faridabad          80
         2 John                         18 Agra               91
         3 Sara                         17 Hisar              86
         4 Rohan                        16 Faridabad          79
         5 Virat                        18 Delhi              80
         6 Komal                        20 Faridabad          95
         7 Preeti                       19 UP                 91

7 rows selected.

SQL>
SQL> -- Disconnect from the database
SQL> DISCONNECT;
```

9. EXCEPTION HANDELING

Aim:

Algorithm:

Coding:

```
-- Create the table
CREATE TABLE geeks (
   g_id NUMBER,
   g_name VARCHAR2(50)
);

-- Insert some data into the table
INSERT INTO geeks (g_id, g_name) VALUES (1, 'GeeksforGeeks');
INSERT INTO geeks (g_id, g_name) VALUES (2, 'Stack Overflow');

-- Commit the transaction
COMMIT;

-- Enable server output
SET SERVEROUTPUT ON;

-- PL/SQL block to select data from the table
DECLARE
   temp VARCHAR2(20);
BEGIN
   SELECT g_id INTO temp FROM geeks WHERE g_name = 'GeeksforGeeks';

EXCEPTION
   WHEN NO_DATA_FOUND THEN
      DBMS_OUTPUT.PUT_LINE('ERROR');
      DBMS_OUTPUT.PUT_LINE('There is no name as');
      DBMS_OUTPUT.PUT_LINE('GeeksforGeeks in geeks table');
END;
/
```

Output:

```
PL/SQL procedure successfully completed.
```

10.TRIGGER

Aim:

Algorithm:

Coding:

```
CREATE TABLE employee_salary(
   EMP_ID number(10),
   SALARY number(10),
   EMP_NAME varchar2(50)
);

CREATE TABLE employee_salary_log(
   EMP_ID number(10),
   NEW_SALARY number(10),
   UPDATED_DATE date,
   UPDATED_BY varchar2(20)
);

CREATE OR REPLACE TRIGGER trg_log_employee_salary
AFTER UPDATE OF SALARY ON employee_salary
FOR EACH ROW
DECLARE
   username varchar2(20);
BEGIN
   SELECT USER INTO username FROM dual;
   INSERT INTO employee_salary_log VALUES (:NEW.EMP_ID, :NEW.SALARY, sysdate, username);
END;
/

INSERT INTO employee_salary VALUES (201, 40000, 'Vikram');
INSERT INTO employee_salary VALUES (301, 35000, 'Nikhil');

-- Corrected update statement with salary value as number instead of string
UPDATE employee_salary SET SALARY = 28000 WHERE emp_id = 101;

-- Commit the changes
COMMIT;
```

:

Output:

```
SQL>
SQL> CREATE OR REPLACE TRIGGER trg_log_employee_salary
  2  AFTER UPDATE OF SALARY ON employee_salary
  3  FOR EACH ROW
  4  DECLARE
  5      username varchar2(20);
  6  BEGIN
  7      SELECT USER INTO username FROM dual;
  8      INSERT INTO employee_salary_log VALUES (:NEW.EMP_ID, :NEW.SALARY, sysdate, username);
  9  END;
 10  /

Trigger created.

SQL>
SQL> INSERT INTO employee_salary VALUES (201, 40000, 'Vikram');

1 row created.

SQL> INSERT INTO employee_salary VALUES (301, 35000, 'Nikhil');

1 row created.

SQL>
SQL> -- Corrected update statement with salary value as number instead of string
SQL> UPDATE employee_salary SET SALARY = 28000 WHERE emp_id = 101;

0 rows updated.

SQL>
SQL> -- Commit the changes
SQL> COMMIT;

Commit complete.
```

11.PRODUCRE

Aim:

Algorithm:

Coding:

```
SET SERVEROUTPUT ON;
DECLARE
   a NUMBER;
   b NUMBER;
   c NUMBER;

   PROCEDURE findMin(x IN NUMBER, y IN NUMBER, z OUT NUMBER) IS
   BEGIN
      IF x < y THEN
         z := x;
      ELSE
         z := y;
      END IF;
   END;
BEGIN
   a := 23;
   b := 45;

   findMin(a, b, c);

   DBMS_OUTPUT.PUT_LINE('Minimum of (23, 45): ' || c);
END;
/
```

Output

```
Minimum of (23, 45): 23

PL/SQL procedure successfully completed.
```